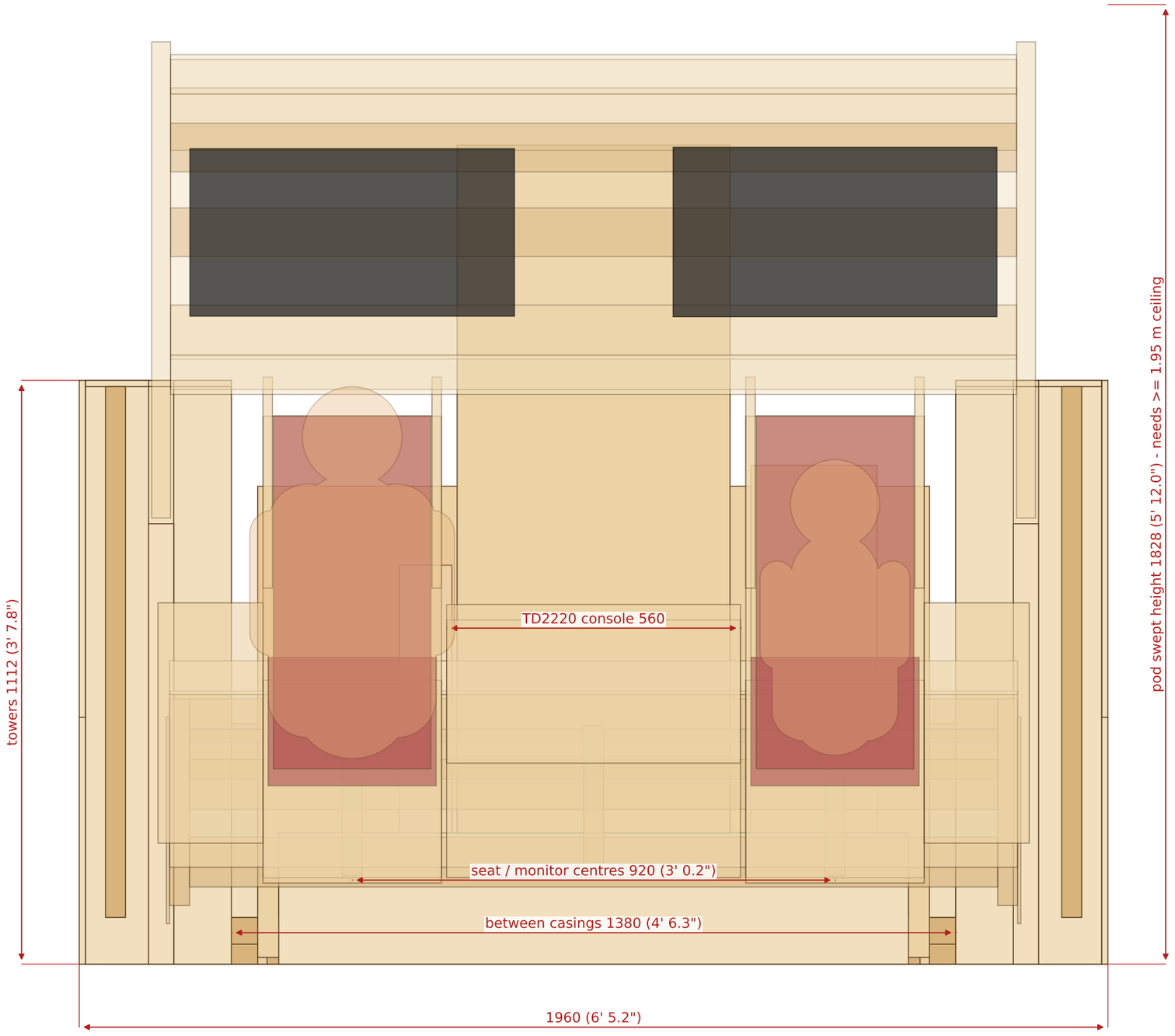
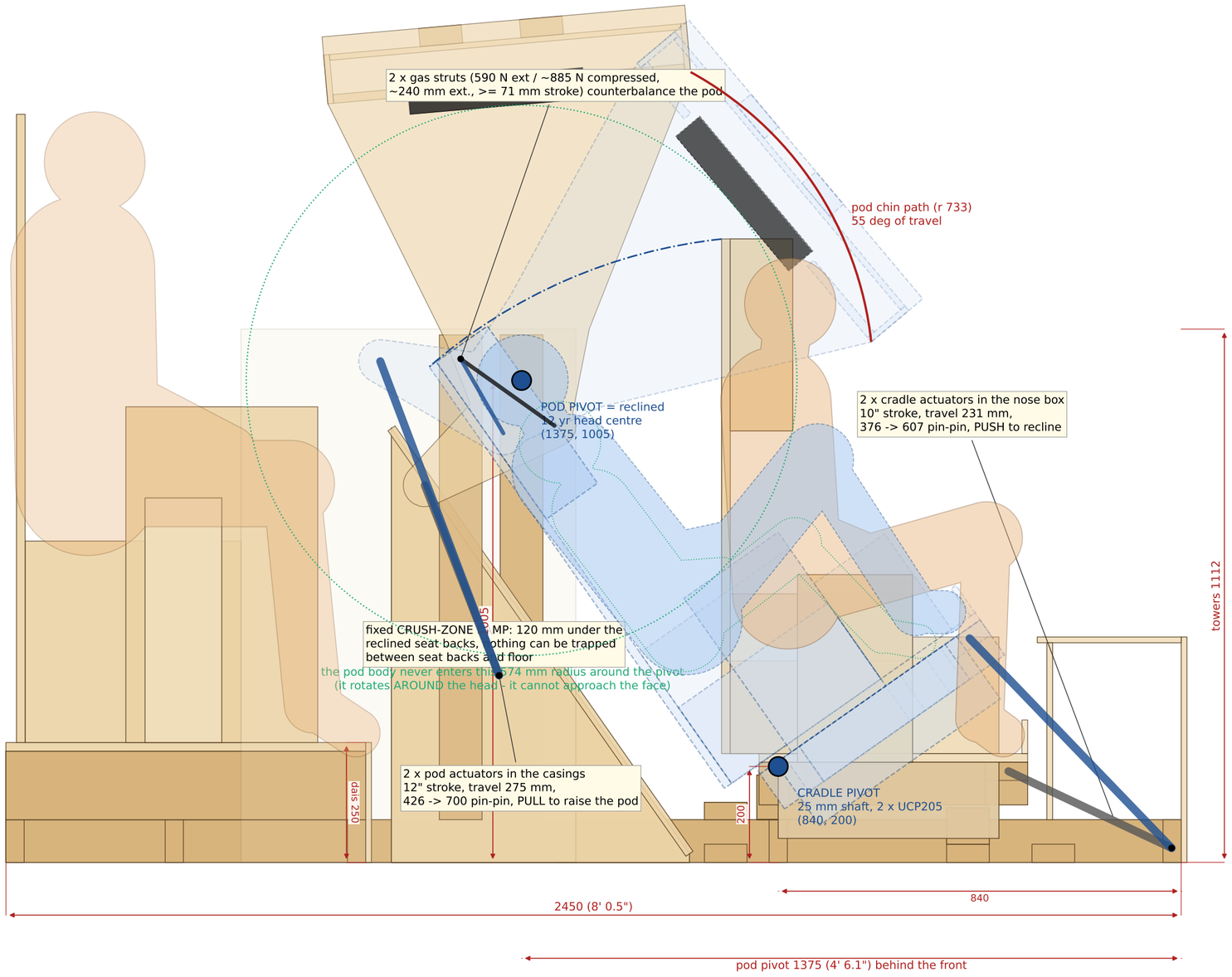
1. Front elevation, FLIGHT position (cradle reclined 35 deg, pod down) - nose box front skin removed

Pod (blue-grey) carries both monitors, the 4 dash panels (lower band) and the 3 overhead panels (top band). The cradle carries the seats, the TD2220 centre console, both side consoles and the sticks. Dad's dais (behind) is fixed.

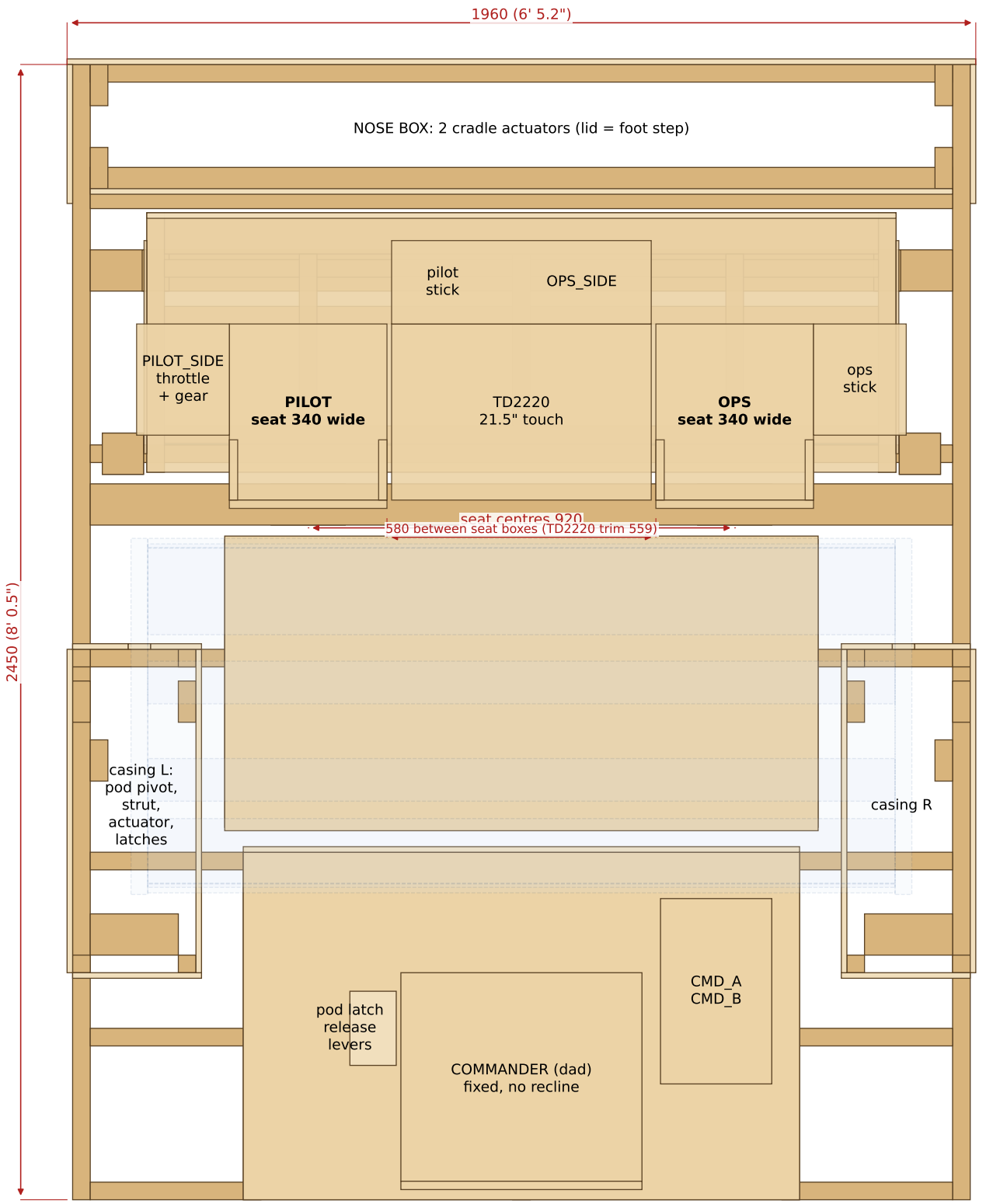


2. Side elevation + MOTION GEOMETRY (from the right, front at the right). Solid = BOARDING (upright, pod up 95 deg), blue dashed = FLIGHT (recline 35 deg, pod down 40 deg)

MOTION: hold FLIGHT -> cradle reclines 0 -> 35 deg (~39 s), then the pod swings down 95 -> 40 deg (~34 s). Hold BOARD -> pod up first, then seats upright.
Clearances at the hard stops (12 yr manikin = worst case): pod chin to thighs/shins 195 mm (ISO 13854 leg >= 180), pod to head 251 mm, pod parked UP: lowest point over the seats z = 1639 mm.
Eye -> screen centre: 12 yr 542 mm, 6 yr 590 mm. Pod mass 44 kg (model), CG 589 mm from its pivot.



3. Plan view, BOARDING position (from above, front at the top; pod parked up shown dashed)



4. Mechanism sizing, locks, hard stops and the hardwired interlock

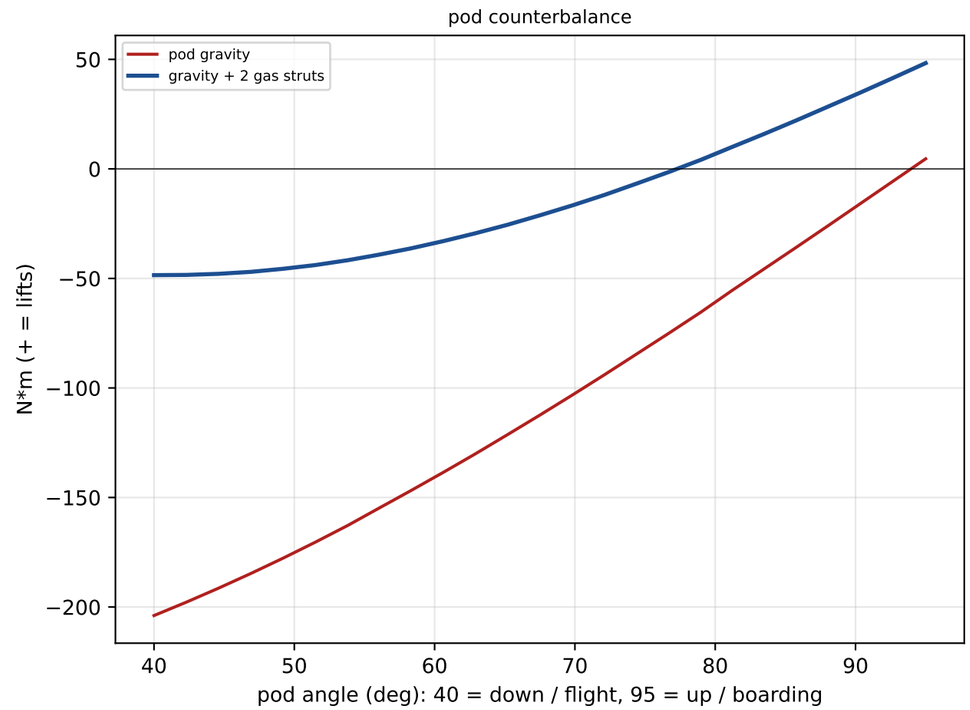
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CRADLE (both kid seats, consoles, TD2220) - recline 0 -> 35 deg about the 25 mm shaft at (840, 200)
  design load: 2 occupants x 50 kg + cradle ~158 kg  -> gravity torque -263 .. +270 N*m
  (negative = wants to sit up: rests on the front rubber pads; positive = wants to lie back: rests on the recline stop blocks)
  2 actuators, 10" (254 mm) stroke, retracted ~364 pin-pin; used travel 376 -> 607 mm
  peak force 775 N EACH (incl. +40 N*m friction) -> BUY 2 x 1500 N (330 lbf) 12 V, self-locking (acme), ~5-8 mm/s

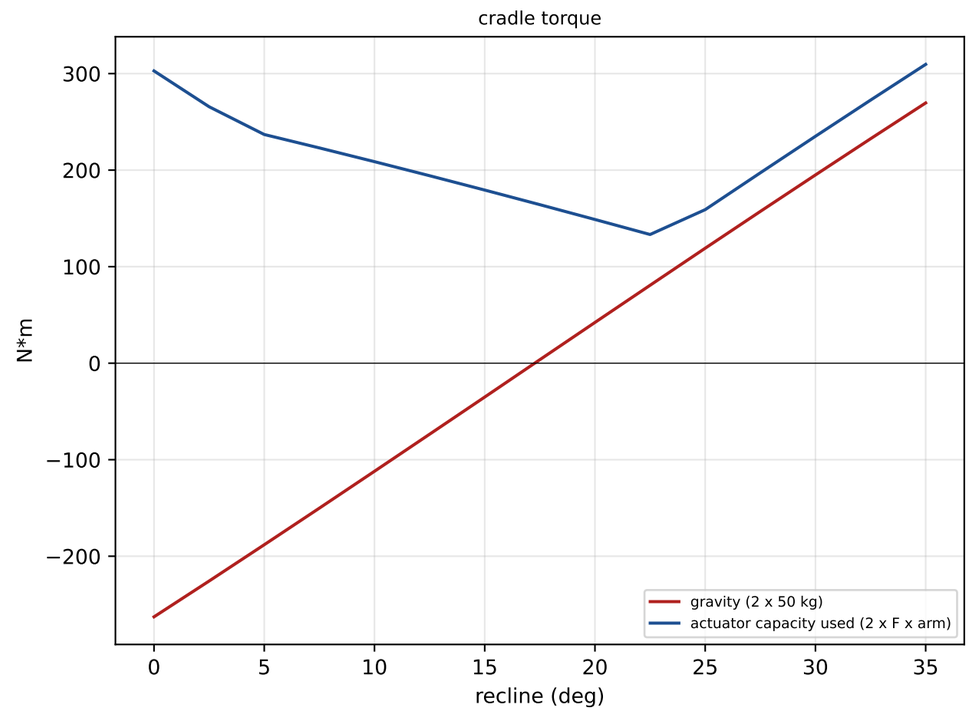
POD (2 monitors, 7 panels, ~44 kg) - swings 95 -> 40 deg about (1375, 1005) = the reclined 12 yr head centre
  gravity torque -204 .. 5 N*m;  with the 2 gas struts: -48 .. +48 N*m
  -> pod rests on the DOWN stop at the bottom and on the UP stop at the top, near-neutral in between
  gas struts: 2 x ~590 N (F1, extended), ~240 mm extended, >= 71 mm stroke, ball studs; adjustable-valve struts let you trim to +-0 N*m
  2 pod actuators, 12" stroke (retracted ~415), travel 426 -> 700 mm, peak 141 N each
  if BOTH struts fail: 390 N each -> BUY 2 x 1000 N (225 lbf), self-locking, slow (<= 10 mm/s => ~27 s)

HARD STOPS (steel, bolted; they - not the actuators - define the end positions):
  cradle upright: rails on 6 mm rubber pads (z 114);  cradle reclined: seat boxes land on rubber stop blocks at 35 deg (+0.5)
  pod DOWN: 6 mm steel stop pin on each cheek crank lands on a rubber-faced steel angle in the casing at 40 deg
  pod UP: same pin on a second stop at 95 deg.  External limit switches stop the actuators 2-3 mm before each stop.

LOCKS (mechanical, both ends):
  pod: spring-loaded steel pawl in each casing drops into a notch in a 6 mm steel latch plate on the cheek at UP and at DOWN
       (ramped like a door latch: it clicks in by itself).  Dad releases both pawls with one Bowden-cable lever LEFT of his seat;
       a micro switch on the lever is in series with the pod motors -> the pod can only move while dad holds the lever
       AND the switch (two-hand hold-to-run).  'Pawl engaged' micro switches feed the interlock below.
  cradle: self-locking actuators + stops; the pod interlock keeps it from moving under a lowered pod.

HARDWIRED INTERLOCK (no microcontroller can move anything; game only READS 2 dry contacts):
  12V+ --[20A fuse]--[MOTION key]--[E-STOP NC]--+--CREW POS (ON)-OFF-(ON) hold-to-run DPDT (drives 4 relays)
  FLIGHT (toggle up):
    cradle RECLINE runs  while  POD_UP_LOCKED (both pawls)  AND  NOT CRADLE_RECLINED limit
    pod DOWN runs        while  CRADLE_RECLINED (both sides) AND  release lever held  AND  NOT POD_DOWN limit
  BOARD (toggle down):
    pod UP runs          while  release lever held  AND  NOT POD_UP limit
    cradle UPRIGHT runs  while  POD_UP limit AND POD_UP_LOCKED  AND  NOT CRADLE_UPRIGHT limit
  => the pod can never come down unless the seats are fully reclined, and the seats never move under a lowered pod.
  Pod chin carries a pressure-sensitive SAFETY EDGE (NC) in series with 'pod DOWN'.
  Limit switches: roller-lever micro switches (NC) on the stops, 1N5408 diode across each so you can always reverse.
  Relays: 4 x 30 A automotive SPDT (Bosch-type) wired as 2 H-bridges; 12 V 20 A supply; 14 AWG motor wire.
```





5. Pinch points, guards and computed clearances

Computed by frame_model.py from the CAD (manikins: 95th pct 12 yr / 50th pct 6 yr, capsule bodies).

POD (sweeping 95 -> 40 deg) vs the RECLINED kids - closest approach anywhere in the travel:

PILOT upper arm	157 mm	(pod at 85.8 deg,	Pod cheek L)
PILOT thigh	195 mm	(pod at 40.0 deg,	Pod chin plate)
PILOT shin	220 mm	(pod at 40.0 deg,	Pod chin plate)
PILOT torso	234 mm	(pod at 85.8 deg,	Pod cheek L)
OPS upper arm	245 mm	(pod at 88.1 deg,	Pod cheek R)
PILOT head	251 mm	(pod at 40.0 deg,	Pod cheek L)
OPS head	261 mm	(pod at 76.7 deg,	Pod cheek R)
OPS torso	291 mm	(pod at 88.1 deg,	Pod cheek R)
PILOT neck	296 mm	(pod at 88.1 deg,	Pod cheek L)
PILOT forearm	306 mm	(pod at 92.7 deg,	Pod cheek L)

CRADLE (reclining with the pod parked up) vs fixed structure / pod / dad:

34 mm	Cradle rail L	<-> Nose box rear wall	at 7.5 deg
35 mm	PILOT kid upper arm	<-> Casing front wall L (inner)	at 32.5 deg
35 mm	PILOT kid upper arm	<-> Casing inner wall L	at 35.0 deg
35 mm	Cradle rail R	<-> Nose box rear wall	at 5.0 deg
40 mm	Cradle toe lip	<-> Nose box rear wall	at 0.0 deg
47 mm	PILOT kid foot	<-> Nose box rear wall	at 0.0 deg
60 mm	Kid seat back PILOT	<-> Casing top L (inner)	at 27.5 deg
60 mm	Headrest wing PILOT out	<-> Casing front wall L (inner)	at 30.0 deg
60 mm	Kid seat back OPS	<-> Casing inner wall R	at 35.0 deg
60 mm	Headrest wing OPS out	<-> Casing inner wall R	at 32.5 deg
64 mm	PILOT kid shin	<-> Nose box lid	at 0.0 deg
81 mm	PILOT kid foot	<-> Nose box lid	at 17.5 deg

POD vs structure:

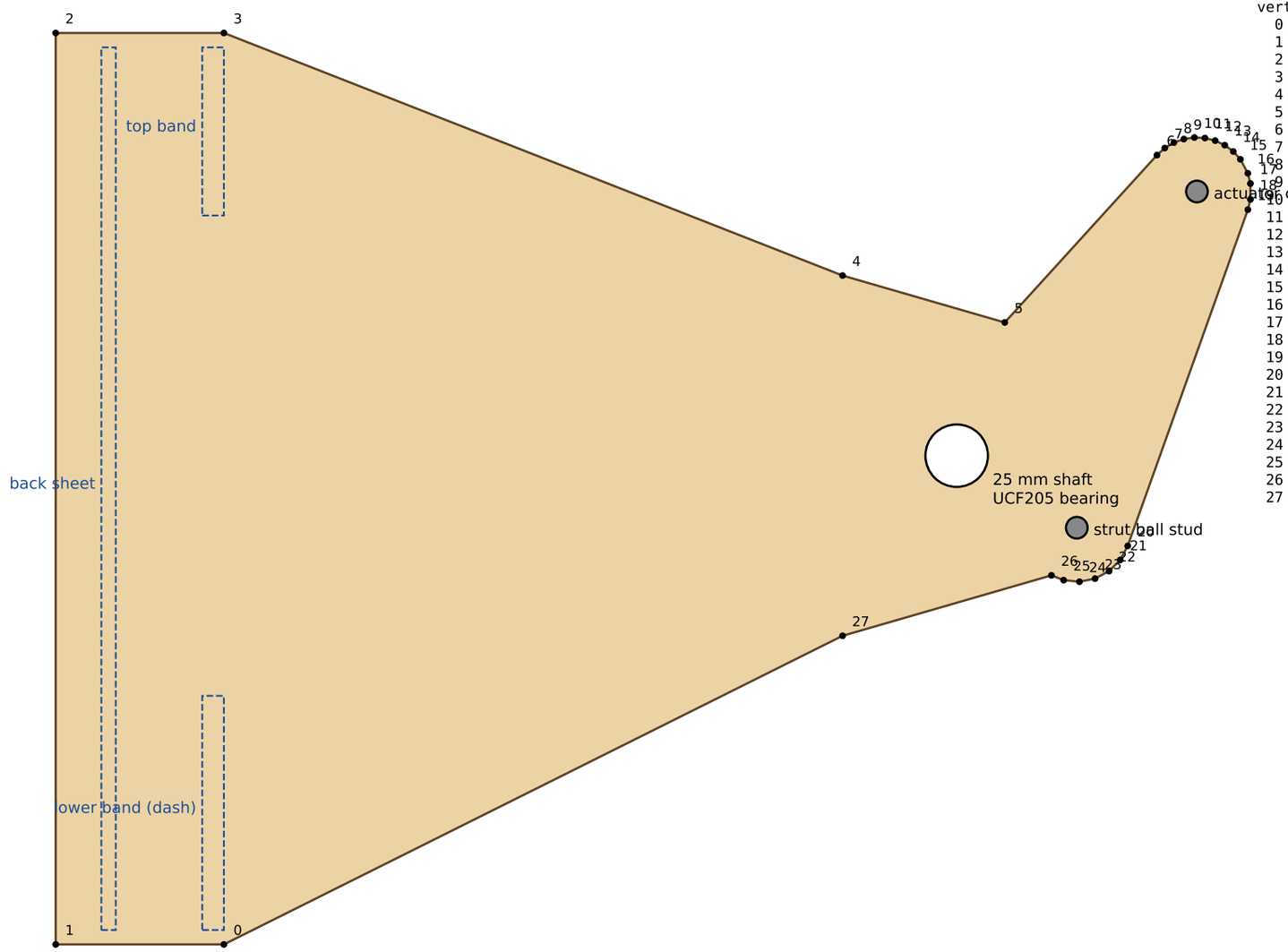
176 mm	Pod cheek L <-> (cradle reclined) Kid seat back PILOT	at 79.0 deg
176 mm	Pod cheek L <-> (cradle reclined) Headrest wing PILOT out	at 90.4 deg

Pod cheek in its casing slot: 66 mm to the inner post, 50 mm to the outer post (strut lives there).

HOW EACH PINCH POINT IS HANDLED:

1	pod chin vs knees/shins/hands	: hard stop keeps >= 180 mm (leg); 25 mm foam + NC safety edge on the chin; slow (<= 10 mm/s)
2	pod vs heads	: pivot = head centre -> constant radius, the pod cannot close on a head (>= 250 mm)
3	cheeks in the casing slots	: 48 mm slot, cheek 36 mm -> 6 mm each side (< finger) + brush strip both edges
4	struts / pod actuators	: fully inside the casings (12 mm ply, screwed lids); only the cheek crosses the slot
5	cradle front vs nose box	: >= 30 mm gap all the way (finger 25) + brush strip on the actuator slots
6	cradle sides vs floor	: 6 mm hardboard skirts on the cradle sides overlap the fixed base skins (telescoping guard)
7	under the reclining seat backs	: fixed crush-zone ramp 120 mm below the reclined backs (arm); foam on top
8	seat backs vs casings	: 60 mm side gap, motion is parallel to the wall (no closing edge)
9	kids' arms vs casing front edge	: arm-to-casing >= 35 mm laterally at full recline; 'hands on lap' + dad watches
10	dad	: fixed seat, nothing moves within his reach except the pod back (>= 400 mm from his head)
11	cables across both pivots	: printed pivot clamps <= 60 mm from the axis + 200 mm U service loop in braided sleeve

6. Pod cheek (2 per side, 18 mm ply, laminated) - cutting template coordinates, pivot at (0,0), screen axis to the left



vertex	r(mm, toward screen)	t(mm, up the screen)
0	610	-407
1	750	-407
2	750	352
3	610	352
4	95	150
5	-40	111
6	-167	250
7	-173	256
8	-181	261
9	-189	264
10	-198	265
11	-207	265
12	-215	262
13	-223	259
14	-230	253
15	-236	247
16	-242	235
17	-245	227
18	-245	213
19	-242	205
20	-142	-75
21	-136	-87
22	-127	-96
23	-115	-102
24	-102	-105
25	-89	-104
26	-79	-100
27	95	-150

Latch plate (6 mm steel, bolted to the cheek, centred on the pivot): notches at the UP and DOWN angles.
Stop pin (12 mm steel) through the crank lobe hits rubber-faced steel angles in the casing at both ends.
Print this sheet 1:1 in tiles or transfer the coordinates; both cheeks of a side are glued + screwed together (36 mm).

7. Panel bay detail - how every printed panel bolts to the plywood bands (40 mm grid, M5 into T-nuts)

